เลขที่นั่งสอบ

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Final Examination of Semester 1 — Year: 2017

Subject: 353152 Electronic Circuits I — Section: 05-06

Date: 7 December 2017 — Time: 13.00-15.00

Name:____________________ ID:__________ Field of Study:______________

Instructions:

1. The examination has 9 pages (including this page), 2 sections (31 questions) and a total score of 90 points.
2. Write all your solutions and answers on this examination sheet.
3. This is a closed book examination.
4. You are not allowed to leave the exam room during the first 1 hour after the beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
6. You are not allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Dictionaries and a calculator are allowed in the examination.
9. Electronic communication devices are NOT allowed in the examination room.
10. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University

Part 1 (50 points) Multiple choices

Instruction: Mark the correct answer for the following questions in **the provided answer sheet.**

1. Which of following transistor operating mode represents the short circuit?
   (A) cut-off (B) saturation (C) active (D) open
2. Which class of power amplifiers that conducts less than 180° of input?
   (A) class A (B) class B (C) class AB (D) class C
3. Which class of power amplifiers having Q-point at the cutoff point?
   (A) class A (B) class B (C) class AB (D) class C
4. How can the class A amplifier efficiency be improved from 25%?
   (A) increase input current. (B) reduce input current.
   (C) add transformer at input side. (D) add transformer at output side.
5. What is the benefit of using Darlington pair and feedback pair transistors for push-pull operation?
   (A) reduce output swing (B) increase output power
   (C) increase input current (D) reduce distortion
6. Which type of following power amplifiers having the highest efficiency?
   (A) class A (B) class B (C) class AB (D) class C
7. What is the cause of crossover distortion in push-pull circuit?
   (A) two transistors do not turn on/off simultaneously.
   (B) heat at transistor.
   (C) two transistors are not the same type (PNP and NPN).
   (D) input current is the sine wave.
8. What is the value of I_{DC} of class B amplifier if peak output current $I_L(p)$ is 10 A.
   (A) 0.318 A (B) 0.636 A (C) 0.707 A (D) 1.414 A
9. What is the value of $P_0(AC)$ of class B amplifier if $V_L(p\text{-}p)$ = 10 V and R_L = 2 ohms.
   (A) 50 W (B) 25 W (C) 6.25 W (D) 5.25 W
10. What is the purpose of using push-pull circuit for class B amplifier?
    (A) to obtain full cycle (B) to reduce the distortion
    (C) to increase speed of signal (D) to reduce the cost of amplifier
11. Which of the following is (are) the application(s) of a transistor?
    (A) Signal protector (B) Voltage regulator
    (C) Computer logic circuitry (D) DC Adapter
12. In an emitter-bias configuration, the _____ the resistance R_E, the _____ the stability factor, and the _____ stable is the system.
    (A) smaller, lower, less (B) larger, more, more
    (C) smaller, more, more (D) larger, lower, more

13. The _____the stability factor, the _____sensitive the network is to variations in that parameter.

(A) higher, more
(B) higher, less
(C) lower, more
(D) None of the above

14. In a transistor-switching network, the operating point switches from _____ to _____ regions along the load line.

(A) cutoff, active
(B) cutoff, saturation
(C) active, saturation
(D) None of the above

15. In a transistor-switching network, the level of the resistance between the collector and emitter is _____ at the saturation and is _____at the cutoff.

(A) low, low
(B) low, high
(C) high, high
(D) high, low

16. Calculate the delay time in a transistor switching network if t_{on} is 56 ns, t_f = 14 ns, and t_r = 20 ns

(A) 70 ns
(B) 42 ns
(C) 36 ns
(D) 34 ns

17. _____is the least stabilized circuit.

(A) Fixed bias
(B) Emitter-stabilized bias
(C) Voltage divider
(D) Voltage feedback

18. _____ is less dependent on the transistor beta.

(A) Fixed bias
(B) Emitter-stabilized bias
(C) Voltage divider
(D) Voltage feedback

19. The emitter resistor in an emitter-stabilized bias circuit appears to be _____ in the base circuit.

(A) larger
(B) smaller
(C) the same
(D) None of the above

20. Changes in temperature will affect the level of _____.

(A) current gain β
(B) leakage current I_{CEO}
(C) both current gain β and leakage current I_{CEO}
(D) None of the above

21. In a fixed-bias circuit, the slope of the dc load line is controlled by _____.

(A) R_B
(B) R_C
(C) V_{CC}
(D) V_{BE}

22. In any amplifier employing a transistor, the collector current I_C is sensitive to _____.

(A) β
(B) V_{BE}
(C) I_{CO}
(D) V_{CE}

23. In a voltage-divider circuit, which one of the stability factors has the least effect on the device at very high temperature?

(A) $S(I_{CO})$ (B) $S(V_{BE})$ (C) $S(\beta)$ (D) Undefined

24. Which of the following voltages must have a negative level (value) in any NPN bias circuit?

(A) V_{BE} (B) V_{CE} (C) V_{BC} (D) None of the above

25. In a collector feedback bias circuit, the current through the collector resistor is _____ and the collector current is _____.

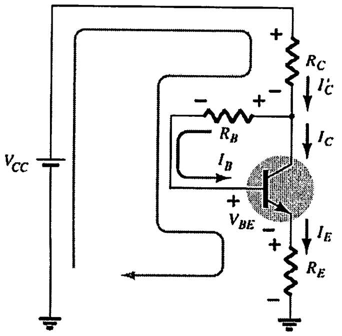
(A) I_C , I_C (B) $I_B + I_C$, I_C

(C) I_B , I_C (D) None of the above

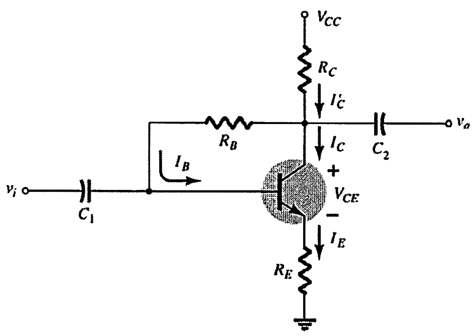
<u>Part 2</u> (40 points) Calculation

Instruction: Show the mathematical expression and answers of following problems.

1. (10 points) From the collector-feedback configuration below, show the mathematical derivation of the current I_B and voltage V_{CE}.

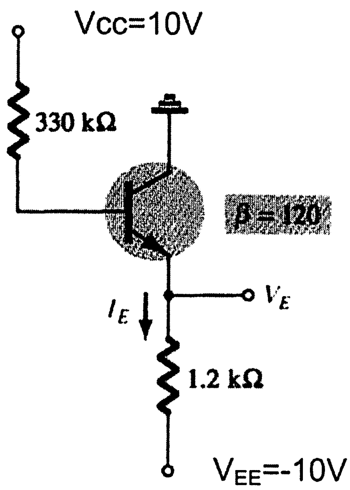


2. (5 points) Given R_C=5kΩ, R_B=200kΩ, R_E=1.5kΩ and V_{CC}=12V, find the current I_B and voltage V_{CE}.

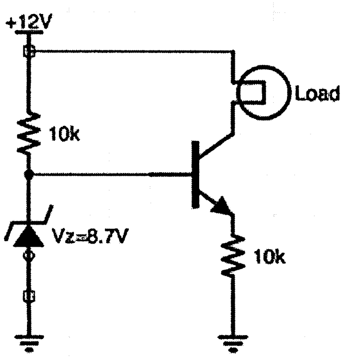


β=90

3. (5 points) Determine the level of V_E , I_B and I_E



4. (5 points) Calculate the constant current I at the load in the transistor/zener constant-current source circuit below. Given $I_C \approx I_E$.



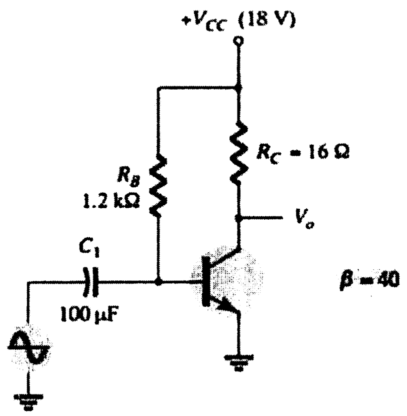
5. (5 points) From the table below, determine the current ΔI_C if the temperature is changed from 25 °C to 100°C. Given R_B = 200 kΩ and R_E = 2 kΩ.

Given:

$$S(I_{CO}) = (\beta + 1)\frac{1+R_B/R_E}{(\beta+1)+R_B/R_E}.\ \text{and}\ S(V_{BE}) = \frac{-\beta}{R_B+(\beta+1)R_E}$$
$$\Delta I_C = S(I_{CO})\Delta I_{CO} + S(V_{BE})\Delta V_{BE}$$

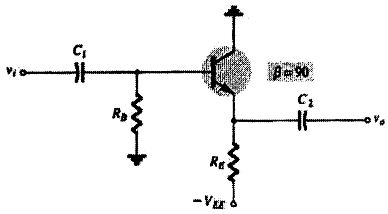
T (°C)	I_{CO} (nA)	β	V_{BE}(V)
−65	0.2×10^{-3}	20	0.85
25	0.1	50	0.65
100	20	80	0.48
175	3.3×10^{3}	120	0.3

6. (10 points) Determine the input and output power of the Class A amplifier circuit below. Given the current of input signal is $7mA_{rms}$.



Handout

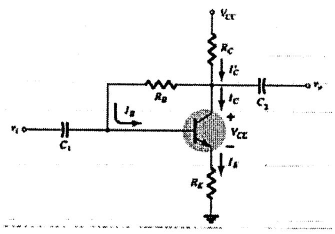
Emitter-follower



$$V_{CE} = V_{EE} - I_E R_E$$

$$I_B = \frac{V_{EE} - V_{BE}}{R_B + (\beta + 1) R_E}$$

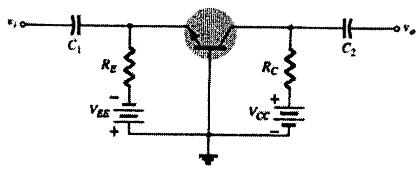
Collector-feedback



$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta (R_C + R_E)}$$

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

Common-base



$$I_E = \frac{V_{EE} - V_{BE}}{R_E}$$

$$V_{CE} = V_{EE} + V_{CC} - I_E (R_C + R_E)$$

Class A amplifier

$$P_i(DC) = \frac{V_{CC}^2}{2R_C}$$

$$P_0(AC) = \frac{V_{CC}^2}{8R_C}$$

Transformer-couple class A amplifier

$$\%h = 50 \left(\frac{V_{CE(\max)} - V_{CE(min)}}{V_{CE(max)} + V_{CE(\min)}} \right)^2$$

$$\frac{R_L'}{R_L} = \frac{R_1}{R_2} = \left(\frac{N_1}{N_2} \right)^2 = a^2$$

Class B amplifier

$$P_0(AC) = \frac{V_L^2(p-p)}{8R_L} = \frac{V_L^2(p)}{2R_L}$$

$$P_i(DC) = V_{CC} I_{dc} = V_{CC} \left(\frac{2}{\pi} I_L(p) \right)$$

$$\%\eta = \frac{P_0(ac)}{P_i(dc)} \times 100\%$$

Answer Sheet

Name:______________________________ ID:________________

Subject: 342152 Electronic Circuits I

	(A)	(B)	(C)	(D)
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